C-M-371G
July 15, 1975
SUPERSEDING
Int.Fed.Spec. C-M-00371F(Army-GL)
June 25, 1974 and
Fed. Spec. C-M-371D
May 13, 1966

FEDERAL SPECIFICATION

MILK, EVAPORATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for evaporated milk for the use by all Federal agencies.

1.2 Classification. Evaporated milk covered by this specification shall be of the following types, as specified (see 6.2):

Types

I - Evaporated milk (stabilized) (see 3.2)
II - Evaporated milk (stabilized) with added vitamin A

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-C-29 - Canned Subsistence Items, Packaging and Packing of
PPP-C-96 - Cans, Metal, 28 Gage and Lighter

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil agencies)

FSC 8910

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-671 - Sanitary Standards for Milk Evaporating and Drying Plants

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

U.S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

Definitions and Standards for Evaporated Milk

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

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U.S. Department of Agriculture

Directory of Meat and Poultry Inspection Program Establishments, and Officials
Brucellosis Eradication - Recommended Uniform Methods and Rules, Agricultural Research Service, U.S. Department of Agriculture, Bulletin ARS 91-79
U. S. Standard for Condition of Food Containers

(Application for copies should be addressed to the U.S. Department of Agriculture, Information Division, Agricultural Marketing Service, Washington, DC 20250.)

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service

Dairy Plants Surveyed and Approved for USDA Grading Service

(Applications for copies should be addressed to the Dairy Division, AMS, U.S. Department of Agriculture, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Offical Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

Association of Vitamin Chemists

Methods of Vitamin Assay (3rd Edition)

(Application for copies should be addressed to Interscience Publishers, a Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, NY 10016.)

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Milk. The raw milk used in the preparation of this product shall comply with all the quality requirements specified in the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service for the manufacture of evaporated milk. The milk shall be drawn from cows in herds accredited as tuberculosis-free and certified brucellosis-free by the U.S. Department of Agriculture, or herds that have passed an annual tuberculosis test and meet USDA requirements for an individually certified herd, or from cows in herds located in:

(1) A Modified Accredited Tuberculosis Area; and
(2) Either (a) A Certified Brucellosis Area; or
(b) Modified Certified Brucellosis Area; or
(3) An area in the process of being certified or accredited by the USDA in accordance with USDA Agricultural Research Service Bulletin 91-76. The milk shall be practically free from colostrum, fresh, wholesome, and normal in appearance and odor and shall be subject to inspection by the procuring agency or duly authorized representative.

3.1.1.1 Bacterial and sediment requirements. The raw milk supply shall comply with the minimum bacterial and sediment requirements for acceptable milk set forth in the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service.

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3.1.2 Stabilizers and emulsifiers. The type and quantity of stabilizers and, when used, emulsifiers, shall be as permitted in the Definitions and Standards of Identity for Evaporated Milk of the Federal Food, Drug, and Cosmetic Act.

3.1.3 Vitamin D. The vitamin D component shall be a concentrate of vitamin D of food grade quality dissolved in a food oil, but if such oil is not milkfat, the quantity added shall be not more than 0.01 percent by weight of the finished product.

3.1.4 Vitamin A. Vitamin A shall be a concentrate of Vitamin A ester (palmitate). The palmitate shall be refined so that when added to the product at the required level (see 3.3.2), it will not impart a fishy or other objectionable odor or flavor to the product.

3.1.5 Milk products. Skim milk, cream, concentrated milk, and concentrated skim milk used for standardization shall be obtained from raw milk complying with 3.1.1.

3.2 Processing. All incoming milk, unless processed within 2 hours, shall be cooled immediately to 45° F (7° C) or lower until start of processing. The product shall be prepared by the evaporation of a portion of the water from whole or standardized milk which meets the requirements of 3.1.1 and 3.1.1.1. The evaporated milk shall be homogenized to delay fat separation and standardized to meet the analytical requirements (see 3.3.2). Stabilizers and stabilizing salts meeting the requirements of 3.1.2 shall be added to both types I and II products. The evaporated milk shall contain added vitamin A, when type II product is specified. The evaporated milk shall be sealed in cans and processed by heat or heat treated and canned aseptically to prevent bacterial spoilage. The cans processed by heat shall be cooled to 100° F (38° C) or lower immediately after processing except that the 6-3/4 pound (3.1 kg) cans shall be cooled to 110° F (43° C) or lower immediately after processing and further cooled to 100° F (38° C) before packing.

3.3 Finished product.

3.3.1 Physical requirements. (See table V for product performance after incubation as applied to aseptic processed and packaged evaporated milk).

3.3.1.1 Flavor. The product shall possess a sweet, pleasing, and desirable flavor and shall be free from scorched, oxidized, or other objectionable tastes and odors.

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3.3.1.2 Body and texture. The product shall be of uniform consistency and appearance. It shall be smooth and free from fat separation, lumps, clots, gel formation, coarse-milk-solids precipitate, sedimentation, and extraneous material.

3.3.1.3 Color. The color shall be of a natural white or light cream.

3.3.1.4 Degree of burn-on. The interior walls of the container shall not show excessive burn-on of product fused to side of can (product fused to more than 75 percent of the inner surface of the can).

3.3.1.5 Keeping quality (applicable only to product aseptically processed and packaged). Samples incubated at 90° F to 100° F (32° C to 38° C) (see 4.5.3) for seven days, shall meet the physical requirements of 3.3.1.1 through 3.3.1.4. All of the samples incubated shall meet the microbiological requirements specified in the definitions given for aseptic processing and packaging in FDA Part 128b of Thermally Processed Low-Acid Foods Packaged in Hermetically Sealed Containers.

3.3.2 Analytical requirements. The product shall conform to the following analytical requirements.

Milkfat (not less than)	7.5 percent
Total milk solids, (not less than)	25.5 percent
Vitamin D	25 International Units (IU) per fluid oz. (30 ml) 1/
Vitamin A (type II only)	125 IU per fluid ounce (30 ml) (minimum) 1/

1/ Additions shall be within the limits of good manufacturing practice.

3.4 Age of product. The product shall be shipped within 60 days after manufacture unless otherwise specified (see 6.2) by the procuring activity.

3.5 Plant qualification. The product shall originate, be processed, handled, packaged and stored in plants which are approved under the USDA General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service, and listed by the U.S. Department of Agriculture in the document titled Dairy Plants Surveyed and Approved for USDA Grading Service, and approved for the finished product described in this specification, or when procurements are for Military use in plants meeting the requirements of MIL-STD-671.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and Regulations Promulgated Thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examinations and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.1.1 USDA certificate of quality and inspection. Unless otherwise specified (see 6.2) the supplier is required to furnish an official inspection certificate issued by the U.S. Department of Agriculture certifying that the quality of the product supplied is as required by the contract or order (see 6.2 for note on small lots).

4.2 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standard for Condition of Food Containers shall apply for the condition inspection of containers. (For applicable criteria governing inspection of containers for the Military see 4.4.6 through 4.4.10).

4.3 Plant qualification conditions. The product furnished under the document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.5.

4.4 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Component and material inspection. In accordance with 4.1 components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Inspection of milk and milk products.

4.4.1.1.1 Animal sources. Inspection shall be made to determine that cows supplying the milk used in the preparation of this product are in herds designated by the U.S. Department of Agriculture as tuberculosis-free and from herds designated brucellosis-free or in the process of being so designated. Determination of herd status regarding tuberculosis and brucellosis shall be made by contacting the nearest office of the Agricultural Research Service of the Animal Health Division. (Location of this office may be ascertained by consulting the publication titled "Directory of Meat and Poultry Inspection Establishments, Circuits and Officials"). Nonconformance to either of the above requirements shall be cause for rejection of the lot.

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4.4.1.1.2 Establishment sources. Examination shall be made to determine that all milk supplies used in the manufacture of the finished product have been handled, processed, packaged and shipped by establishments approved as Military sources of supply as listed in the Directory of Sanitarily Approved Food Establishments for Armed Forces Procurement or listed by the U.S. Department of Agriculture in the document entitled: Dairy Plants Surveyed and Approved for USDA Grading Service and approved for the product to be used in the manufacture of the finished product specified in this document. Nonconformance to the aforementioned establishment sources shall be cause for rejection of the milk supply or the involved quantity of finished product made therefrom, as applicable.

4.4.1.1.3 Examination of raw whole milk. Examination and testing of raw whole milk shall be made to determine compliance with the requirements of 3.1.1 as concerns animal health, appearance and odor and with the referenced requirements of 3.1.1.1 as concerns bacterial and sediment content. Records of all required examinations and tests shall be maintained and made available for review. Failure to comply with one or more of the above referenced requirements, reflected by actual examination and tests or by examination of records, shall be cause for rejection of the involved lot of finished product.

4.4.1.2 Stabilizers and emulsifiers. Acceptance of these ingredients for compliance with the requirements of 3.1.2 shall be based on the examination of the labels, invoices, and similar documents. Nonconformance to the requirements of 3.1.2 shall be cause for rejection of the involved finished product.

4.4.2 In-process examination.

4.4.2.1 Equipment, procedures, and controls. Examination shall be performed, during processing of the milk, to determine compliance with 3.2 pertaining to equipment, procedures, and controls. Records shall be maintained for specific procedures of time and temperature controls covering product included in the purchase contract. Noncompliance with the requirements of 3.2, as determined by examination of records or actual examination, shall be cause for rejection of the involved finished product.

4.4.2.2 Formulation. Examination shall be made during processing of the product to determine conformance to 3.1.2, 3.1.3 and 3.1.4 with regard to limitation of the ingredients in the finished product. Formulation records shall be maintained. Noncompliance, as determined by examination of records or actual examination, shall be cause for rejection of the involved finished product.

4.4.3 For Military procurement (testing of can and lid). Testing of the tinplate on can or can with lid shall be as shown below. Requirement of test characteristic is applicable to lot average. Product shall be rejected if test results indicate nonconformance to test requirements.

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Component 1/	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported as	Test ref. and rqmt. para.
Can or can with lid	Cans or cans with lids	Cans or cans with lids	S-2	Tinplate	Nearest 0.01 lb/base box	4.5.2 5.1.1

1/ Testing of can coating when required shall be on the same cans used for testing of tinplate.

4.4.4 Finished product inspection.

4.4.4.1 For civil agencies.

4.4.4.1 Lot size, sample unit and acceptable quality levels. An inspection lot shall be one car lot or less converted to number of primary containers of evaporated milk to be inspected at one place and at one time. More than one production code may be included in the lot if the number of cases from each code represented in the lot is known and the production dates are all within fifteen days of one another. The sample unit for examinations required in tables II, III and VI shall be one primary container and for tables IV and V shall be the contents of one primary container. Sampling plans shall be in accordance with table IA. The acceptable quality levels (AQL) shall be expressed in defects per hundred units.

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TABLE IA. Single sampling plans (Civil agencies)

Number of primary containers in the inspection lot	All lots-table II			All lots-tables III and IV					Only applicable to aseptically canned lots					
	Minimum sample size	Total Ac	Re	Minimum sample size	Major Ac	Re	Total Ac	Re	Minimum sample size	Table V Ac	Re	Minimum sample size	Table VI Ac	Re
5-500	9	0	1	6	1/	1/	1	2	21	0	1	5	0	1
501-35,000	13	1	2	13	0	1	2	3	21	0	1	5	0	1
over 35,000	29	2	3	29	1	2	4	5	48	1	2	11	1	2

1/ Lot with one or more defect shall be rejected.

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4.4.4.1.2 Examination of containers, labeling and marking. Examination of the exterior condition of the primary containers, labels and marking shall be in accordance with 4.2. (See purchase contract for type and size, primary and shipping container and for any special labeling and marking requirements.)

4.4.4.2 For Military agencies.

4.4.4.2.1 Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise stated herein.

4.4.4.2.2 Examination of end item. Classification of defects found during examination shall be in accordance with tables II, IV and V and 4.4.6 through 4.4.11. The sampling and acceptance criteria of the above referenced tables shall be as shown in table IB. AQL's shall be expressed as defects per hundred units (DHUs) for tables II, IV and V.

TABLE IB. End item examination criteria (Military agencies)

Table	Inspection level	Sample unit	Lot size expressed in	AQL	
				Major	Minor
II	S-3	One primary container	Primary containers	-	2.5
IV	S-2	Contents of one primary container	Primary containers	1.0	4.0
V	S-2	One sealed can	Primary containers	1.0	-

4.4.5 Examination of the net weight, interior packaging characteristics, product characteristics, leakage, and incubation tests.

4.4.5.1 For Civil agencies. Examination shall be in accordance with the following procedure: The appropriate minimum sample size designated in table IA shall be formed by drawing one sample unit per shipping case selected proportionately from the production codes represented in the inspection lot. The applicable acceptance (Ac) and rejection (Re) numbers are given in table IA. Classification of defects found during the examination shall be in accordance with tables II through VI.

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TABLE II. Examination for net weight (for Civil and Military agencies) 1/ 2/

Category	
201	Net weight less than 14-1/4 ounces (404 g) or 6 pounds 11-5/8 ounces (3.05 kg) as applicable

1/ Lot shall be rejected if sample date indicates lot average net weight is less than specified weight. Report to nearest 1/4 ounce or nearest gram.

2/ To facilitate evaluation of net contents of container, net weights shall be obtained for the product. Thirteen fluid ounces equals 14-1/2 ounces net weight.

TABLE III. Examination for interior packaging characteristics (for Civil agencies)

Category	Defect	Standard evaporated can		Round, metal open-top style can	
		Major	Minor	Major	Minor
Interior of container					
101	Interior rusted	X		X	
Interior coating can					
102	(a) Absent	X		X	
103	(b) Blistering area which can be peeled by fingertip abrasion	X			
Enamel or tinplate					
104	(a) Denuded areas other than scratches	X		X	
203	(b) Scratches through enamel or tinplate		X		X

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TABLE IV. Examination for product characteristics (for Civil and Military agencies) 1/

Category	Defects	Major	Minor
Flavor			
101	Slightly scorched	X	
102	Slightly oxidized	X	
103	Very slightly rancid	X	
104	Other objectionable or noncharacteristic odors	X	
Body and texture:			
105	Coarse-milk-solids precipitate or sedimentation	X	
106	Fat separation	X	
107	Lumps, clots, clabbering, gel formation, not uniform consistency	X	
Color			
201	Other than a natural white or light cream color		X
Burn-on			
202	Excessive burn-on (product fused to more than 75 percent of the inner surface of the can)		X

1/ Presence of foreign materials shall be the basis for the rejection of the lot.

TABLE V. Examination of incubation test (for Civil and Military agencies) 1/

Category	Defects	Major
101	Sweller	X
102	Presence of coagulum or lumps	X
103	Bitter, putrid, or other objectional flavor or odor	X

1/ Only applicable to product heat treated prior to aseptic canning; not applicable to product heat treated after canning, see 4.5.3 for procedure

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TABLE VI. Examination for leakage (for Civil agencies) 1/

Category	Defect
151	Leakage

1/ See 4.5.4 for procedure.

4.4.6 Examination of cans (Military agencies). Examination of cans shall be in accordance with the applicable examination criteria of PPP-C-29.

4.4.7 Examination of can labeling (Military agencies). When can labeling is required to be in accordance with MIL-L-1497, examination of cans for labeling shall be in accordance with examination criteria of that specification.

4.4.8 Examination of unit loads (Military agencies). Unit loads shall be examined in accordance with the pertinent examination criteria specified in MIL-L-35078.

4.4.9 Examination of can coating. When exterior coating of cans is required to be in accordance with TT-C-495, examination of can coating shall be in accordance with that specification.

4.4.10 Examination of shipping containers for levels A and B. When boxes are required to be in accordance with PPP-B-636 the examination for assembly, closure and reinforcement shall be in accordance with that specification (see 5.2). In addition, the following defect shall be included in the table for examination; Major: Markings missing, incorrect, or illegible. For level C pack, examination shall be in accordance with the aforementioned specification except that only the defect for marking and the following defect shall apply when applicable: Minor: Not a snug fit.

4.4.11 Examination for age of product at time of shipment. Unless otherwise specified (see 6.2), the product shall be examined to determine compliance with requirements of 3.4. Noncompliance with above requirement as indicated by marked date of pack or examination of code or record, shall be cause for rejection of the lot.

4.4.12 Sampling procedures and acceptance criteria for testing of finished product.

4.4.12.1 For Civil agencies, sampling procedure and acceptance criteria for testing of finished product. The finished product shall be tested for compliance with analytical requirements of 3.3.2 for milkfat and total solids. Procedures for testing shall be in accordance with 4.5.

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4.4.12.1.1 Supplier testing. A representative sample of each production code (one batch or standardized tank), shall be tested. Records of the results shall be kept and be available for review by the Government.

4.4.12.1.2 Sampling for verification testing. The sample size shall be the number of primary containers indicated for "All lots" in table IA but not more than 13 containers.

4.4.12.1.3 Compositing for verification testing. Composite samples, if used, shall be made just prior to testing. Noncompliance with one or more analytical requirements shall be cause for rejection of the lot.

4.4.12.2 For Military agencies. The finished product shall be tested for compliance with the analytical requirements specified in 3.3.2. Procedures for testing shall be in accordance with 4.5. The sample unit for testing shall be an approximate 1-quart (946 ml) composite made up just prior to testing and derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of primary containers. If the test for vitamin A (type II only) and vitamin D made on the initial production lot complies with the requirement, thereafter, the testing for vitamins A and D need to be run every 6 months or when special testing is deemed necessary by the Government. If the initial production lot does not comply with the vitamin D requirement, testing of subsequent lots shall be continued until two successive lots are in compliance. Each result obtained shall be reported to the same decimal or unit as specified in the pertinent requirements, except that vitamin D content shall be reported to the nearest 50 IU (when converted to quart or cc basis). One or more test failures for analytical characteristics shall be cause for rejection of the lot.

4.5 Tests. Test procedures and processing controls which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the procuring agency activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.5.1 Chemical analyses. Chemical analyses, if required by the purchaser, shall be made in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists (except for vitamin A):

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Test	Method	Chapter and section
Milkfat Total solids	Roese-Gottlieb Total solids (method I)	Dairy Products - Section: Evaporated Milk (Unsweetened)
Vitamin D	Vitamin D in milk	Vitamins and Other Nutrients - Section: Bioassay Method
Vitamin A	Methods of Vitamin Assay, 3rd Edition, Association of Vitamin Chemists	Colorimetric (Carr-Price Blue Color)

4.5.2 Tin coating test. When tin coating weight tests are required they shall be determined by any method specified in PPP-C-96.

4.5.3 Incubation test. A complete set of filled and sealed cans (see table IA and IB for sample size), shall be incubated at two temperatures of 90° to 100° F (32° to 38° C) and at 131° F (55° C) for seven days and examined for the defects listed in table V.

4.5.4 Leakage test. The filled and sealed can shall be submerged in water contained in a dessicator or other suitable container while maintaining a vacuum of 10 inches of mercury (atmospheric pressure 29.9 inches) (67.2 kPa) for at least 30 seconds. A leak is indicated by a steady progression of bubbles from the can. Isolated bubbles caused by entrapped air are not considered leakage.

4.5.5 Sediment and bacteriological analyses of raw milk. Sediment and bacteriological analyses of raw milk shall be in accordance with 3.1.1.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.2).

5.1.1 Level A. Not less than 13 fluid ounces (384 ml) (14-1/2 ounces (411 g) net weight) of product shall be filled into a size 300 by 314-315 soldered-on end style, round, metal can with soldered side seam and end seams. The can top shall be pierced with a standard size evaporated milk can vent hole opening and closed by tipping with solder. The can shall be made throughout from either commercial 0.75 pound per base box electrolytic tinplate or commercial 0.75/0.25 pound per base box differential electrolytic tinplate (see 4.5.2), and the can shall be coated overall on the outside with a coating

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conforming to type I or when specified (see 6.2), type III of TT-C-495. The outside length of the can end flange shall be not less than 0.15 inch (4 mm) and shall be flared outward from the body of the can so as to provide an adequate solder bond. Solder used shall be a composition which will produce an adequate solder bond. The can shall be hermetically sealed and shall not leak when tested in accordance with 4.5.4. The can shall be free from defects such as severe dents, improper closure, swells, springers, severe paneling, and pitted rust.

5.1.2 Level C. Not less than 13 fluid ounces (384 ml) (14-1/2 ounces (411 g) net weight) or 97 fluid ounces (2.86 liters) (6 pounds 12 ounces (3.1 kg) net weight) of the product shall be filled, respectively, into a size 300 by 314-315 or 603 by 700 commercial metal can normally used for the product. Commercial packaging which will afford adequate protection to the product will be acceptable. No material shall be used in the packaging of this product which will impart an odor or flavor to the product. All containers shall be sound and clean.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.2). Can size, quantity and arrangement shall be in accordance with table VII.

TABLE VII. Can sizes, quantities and arrangement in shipping containers

Can Size	Quantity	Arrangement		
		Length	Width	Depth
300 by 314-315	48	6	4	2
603 by 700	6	3	2	1

5.2.1 Level A. The shipping container shall be a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC, V2s of PPP-B-636.

5.2.2 Level B. The shipping container shall be a snug-fitting fiberboard box constructed, closed and reinforced in accordance with style RSC, V3c, V3s or V4s of PPP-B-636. Style RSC boxes may be top-opening or end-opening.

5.2.2.1 When specified (see 6.2 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands may be used, one lengthwise and one girthwise.

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5.2.3 Level C. The product, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2.3.1 Civil agencies. When specified (see 6.2), the product, packaged as specified in 5.1, shall be packed into a snug-fitting fiberboard box, constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure of PPP-B-636.

5.3 Unit load (Military agencies). When specified (see 6.2), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Civil agencies.

5.4.1.1 Unit containers. Any commercial labeling or additional labeling as specified that complies with Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder, will be acceptable.

5.4.1.2 Shipping containers. Unless otherwise specified, shipping containers shall be marked in accordance with FED-STD-123 and such other pertinent information as specified by individual agencies.

5.4.2 Military agencies.

5.4.2.1 Cans. Cans shall be labeled in accordance with MIL-L-1497.

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.2.3 Unit loads. Unit loads shall be marked in accordance with MIL-STD-129 and MIL-L-35078.

6. NOTES

6.1 Intended use and technical information. The product(s) covered by this specification is intended for use in cooking and as a substitute for fresh milk (when blended with equal parts of water), if the consumer does not object to a cooked flavor which may vary in degree somewhat between processors or methods of processing. The product, because of its concentration and sterility, offers special advantages to the user:

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a. Convenience, reduced bulk and versatility - it may be kept on hand at ambient temperatures along with other packaged and canned foods. (Note: The product must be refrigerated after the container has been opened.)

b. Substitute for cream or coffee and tea whitener, can be used in cooked cereals, mashed potatoes, vegetables, sauces, creamed dishes and casseroles, as a "binder" for meat loaf mixtures, as a dip for foods to be coated with crumbs for oven or pan-frying.

c. In diluted form, for infant feedings; in flavored milk drinks and when chilled icy cold in undiluted form, it can be whipped for use as a topping.

d. Useful in bivouac situations.

The product requires no refrigeration prior to opening the container but for optimal keeping quality it should be stored at temperatures of 40° to 60° F (4° to 16° C) or below, but should not be frozen (will freeze at approximately 25° F (-4° C). As aging progresses, an age-darkening or browning effect takes place, accompanied by a slight odor suggesting staleness. The cooked flavor appears to intensify with age and high storage temperatures. Stabilized evaporated milk produced under good commercial manufacturing practices will begin to show signs of color and flavor deterioration after 18 - 24 months storage at temperatures below 60° F (16° C). Product held above 70° F (21° C) will begin to show signs of more rapid deterioration. Under these unfavorable storage conditions, a maximum shelf life of from 6 to 12 months may be expected. For long term storage, humidity should be kept at 50% R.H. or below to inhibit rusting of metal cans and label deterioration. If prolonged storage is anticipated (over one year), it may be necessary to invert cases of the product, routinely, to minimize fat and serum separation. A practical inventory control approach is to examine representative samples of each lot and turn the cases whose can contents show a small, soft cream layer. If cans are not turned while cream layer is soft, it becomes firm, then tough, making the dispersion of fat difficult, if not impossible. A recommended can or case inversion schedule after a year's storage at below 60° F (16° C) is as follows:

Invert at 6-month intervals
Initial and continued storage at approximately 70° F (21° C): Invert at 4-month intervals
Initial and continue storage at 80° F (27° C) or above: Invert at 2-month intervals

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In any event, product examination should be made to determine condition and acceptability. (Note: If soft cream plug separation has occurred, it can usually be corrected by vigorous shaking of the container prior to opening.)

6.2 Ordering data. Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of product required (see 1.2).
(c) Sixty-day age limit may be waived unless delayed usage or unfavorable storage conditions are expected (see 3.4).
(d) When USDA certificate of quality and condition are required (see 4.1.1).
(e) When verification testing for analytical, in-process, and component examinations may be waived on small lot orders (see 4.1.1),
(f) When examination for age of product is not required (see 4.4.11).
(g) Level of packaging and packing required (see 5.1 and 5.2).
(h) Amount of product, type, and capacity of container required (see 5.1.1 and 5.1.2).
(i) When coating shall conform to type III of TT-C-495 (see 5.1.1).
(j) When packing specified in 5.2.2.1 or 5.2.3.1 is required.
(k) Type and class of unit load when unit loading is specified (see 5.3).

6.3 Packing specified in 5.2.2.1 is intended for transfer at sea operations or specific overseas operations.

6.4 Destination inspection (Civil agencies). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition, identity and quantity only.

6.5 Military agencies. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.14D/DSAR 4145.7.

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Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP-SS

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

USDA-AMS
GSA-FSS
HEW-FDA
HEW-NIH
VA-DMS

Project No. 8910-0363

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